A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 33

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** Each regional population has its own unique gene pool / DNA profile **1** DNA from seized timber or ivory is compared with the database of regional populations **1** This identifies where the material came from, showing whether it was legally harvested and where poaching occurs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** Radio / GPS transmitters use radio waves - work in air and fresh water, giving positions (GPS stores or transmits location) **1** Radio waves do not travel through seawater so acoustic transmitters (sound) are used for marine animals, giving real-time positions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round.	2

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2** Drones / ROVs, AUVs (eg Autosub), balloons, aircraft, satellites, animals (tags) **up to 2 Drones:** cheap, low disturbance, high-resolution images of areas hard to reach (cliffs, canopy, ice), repeated surveys – infrared cameras detect animals at night **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Cameras record automatically (visible light by day, infrared at night) without researchers being present **1** So animals behave naturally and nests are not deserted – monitoring continues 24 hours **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infrared** → heat radiation / long-wave radiation re-emitted by the Earth	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-406; AQA spec 3.7.2.3.	
02.1	**1 mark each** 1 mark for each of the 4 points below. **1** Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish **1** Species occupy different niches / trophic levels so they do not compete and all food sources are used **1** Bottom feeders re-suspend nutrients boosting plankton / plant growth **1** Predators control over-breeding species (tilapia) so remaining fish grow larger – total yield exceeds any single species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
02.2	**1 mark each** 1 mark for each of the 4 points below. **1 IMTA:** fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) **1** Wastes of one species become nutrients for another so less pollution and extra harvests **1 Aquaponics:** aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) **1** Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish) – rice-fish paddies similar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages **1** Production still depends on (and removes juveniles from) the wild population – carnivores need large fish inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2** Normal diagonal mesh closes as the net fills / stretches so small fish cannot escape – square-mesh panels do not close so juveniles escape to grow and breed – TEDs let turtles out – larger mesh sizes generally **2** Minimum size lets fish reach breeding age before capture (eg UK lobster minimum carapace above breeding size, cod 35 cm) – maximum size / V-notching egg-carrying females protects large, highly fecund breeding adults (Hebridean lobsters over 145 mm, Nassau grouper) **2** NTZs (Lundy, Lamlash Bay, Chagos, Great Barrier Reef) protect breeding populations and habitats – surplus young spill over and increase surrounding catches – damaged communities recover **2** Closed seasons, ideally in the breeding season, let fish grow and breed – fishermen may oppose because shoaling breeders are easy to catch **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	8
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Pelagic species form single-species shoals so fishing stops when the quota is reached **1** In mixed fisheries many species are caught together – once one species' quota is reached it must be discarded (dead) while fishing for others continues **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **quota** → a legal limit on the mass or number that may be caught **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
04.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Impact depends on total population x per capita impact **1** With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added **1** Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	3
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Urban areas are supported by resources brought in from elsewhere – when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage **1** Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
04.4	**Marks shared** Up to 4 marks in total - share them between the 3 points below according to how completely each one is made. **up to 2 Climate:** warming increases evaporation, cloud and albedo – higher CO2 increases photosynthesis and sequestration – hydrological cycle: more evaporation gives more precipitation – populations: density-dependent factors return numbers to carrying capacity **up to 2** In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) **up to 2** Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back • **carrying capacity** → the largest population the area can support **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
05.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive **2** Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light – diversity falls **2** Hedgerow removal destroys habitat / corridors and nesting sites – loss of natural pest predator habitats increases pest damage – larger fields increase wind erosion – genetic contamination and introduced species also accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows **1** These communities depend on the farming continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2

Do not write outside the box

Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** CO_2 from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery – ploughing increases soil respiration **1** Methane from livestock intestines and rice paddies – nitrous oxide from manure and nitrogen fertilisers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per mm^2) **1** They use the polyp's CO_2 and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently **1** They supply the polyp with glucose and oxygen - over 95% of its food energy – the polyp provides a protected habitat and nutrients **1** Polyps also trap plankton with cilia and stinging nematocysts – surplus CO_2 is used to build the calcium carbonate skeleton **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	4
06.2	**1 mark each** 1 mark for each of the 3 points below. **1** High light (shallow, clear water) for the algae – warm stable temperatures (about 18-30 °C) – low turbidity – constant salinity – well-oxygenated water **1** Found only where all these coincide (about 1% of ocean area) **1** Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **range of tolerance** → the range of conditions in which a species can survive	3

Do not write outside the box

Question	Marking guidance	Marks
06.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Fisheries:** support over a quarter of marine species including fish, lobsters, conch **1 Erosion protection:** complex structure dissipates wave energy protecting low coral islands – coral sand builds shores **1 Medicinal discoveries:** AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts **1 Climate control:** carbonate skeletons sequester CO2 **1** Tourism (diving, snorkelling) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4.	
07.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2 Light:** selective felling to create clearings for plants needing high light while retaining dense canopy for shade species – clearings also give warm sand / soil for lizard eggs – leave dead wood – woodland floor plants (bluebells) grow before the canopy leafs **2 Dissolved oxygen:** narrowing a slow river increases flow and turbulence so more oxygen dissolves, suiting trout and mayfly larvae – shallow water warms quickly for egg development **2 Nutrients:** removing fertile topsoil / not fertilising allows plants of nutrient-poor soils (eg sundew on acidic nitrate-deficient soils) that cannot compete on fertile ground – dung patches add local nutrients **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	6

Do not write outside the box

Question	Marking guidance	Marks
07.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Food:** wildflower grasslands for seed-eating birds – providing food species **1 Control of predation:** island nest sites unreachable by foxes – trapping / culling / eradicating introduced rats (Scilly Isles - Manx shearwater and storm petrel breeding success increased) **1 Control of competitors:** removing rhododendron so wildflowers get light **1 Pollination:** a range of flowering plants with different flowering times so pollinators have food all season **1** Seed dispersal species (hippos, elephants, monkeys) conserved – control of introduced pathogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biotic** → living factors / effects of other organisms **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 43-46; AQA spec 3.1.2.3.4, 3.1.3.1.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 C:** Waste products that are the raw materials for other processes	1
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) **1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO_2 **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Incineration uses high temperatures / fuel and releases pollutants – nutrients are not returned as usable raw materials **1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **incineration** → burning waste, usually with energy recovery • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** As sea otters declined (1960-1980) sea urchins increased (to 350) and kelp area fell (to 25) **1** As otters recovered after 1990 urchins declined and kelp recovered **1** Otter and kelp trends are positively correlated – urchins negatively correlated with both **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
09.2	**1 mark each** 1 mark for each of the 3 points below. **1** Sea otters are predators of sea urchins so fewer otters allowed the urchin population to rise **1** Sea urchins graze kelp so the kelp forest was over-grazed and destroyed **1** Kelp forests are habitat / food for many species including otters (keystone species / trophic cascade) – otter recovery restored the balance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **keystone species** → a species whose loss changes the whole community **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

Do not write outside the box

Question	Marking guidance	Marks
09.3	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Over-fishing of sandeels reduced puffins **1** Over-collection of turtle eggs reduced turtles so jellyfish (their food) increased – over-fishing of sardines / herring (competitors) also increased jellyfish **1** Loss of pollinating bees (pesticides, wild flower loss) reduces plant reproduction – loss of elephants / large herbivores reduces seed dispersal of trees **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1 They control abiotic features:** beaver dams create ponds colonised by aquatic species – elephants create clearings and keep water holes open **1** Species that depend on these features (light, water, minerals) cannot survive in their absence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 23-25; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Resources:** wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food – new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • **Biomimetics:** vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples) – most species unstudied - argument for habitat conservation • Physiological research using wildlife species – wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • **Genetic resources:** crop wild relatives, Vavilov centres, gene pools • **Ecosystem services:** atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments – the unknown value of species not yet investigated (precautionary principle) • **Evaluation:** which arguments are most persuasive to decision makers – economic valuation vs intrinsic value – judgement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely	9
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, *Encarsia*, *Cactoblastis*, *Bacillus thuringiensis* - and crop wild relatives in Vavilov centres hold the genes future crops will need.	
	Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness.	
	Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know.	
	Weaker answer - Level 1 (3 marks)	
	Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future.	
	What makes the difference	
	• Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category.	
	• Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits.	
	• 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not.	
10.2	1 mark for the point below. **1** The variety and abundance of life (species, genes, habitats) in an area / on Earth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Setting priorities:** IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • **Legislation and protocols:** Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • **Captive breeding and release:** selection, difficulties, techniques, release site criteria, hard / soft release, monitoring – seed banks • **Habitat conservation:** creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • **Evaluation:** examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers) – cost, politics, climate change – judgement on the most effective combination	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	
	Model answer - Level 5 (23 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Biodiversity is lost through direct exploitation, habitat destruction, introduced species, pollution and changes to abiotic factors, so its conservation needs methods that address each cause, at scales from a single species to international law. Their effectiveness varies with the threat, the species and the resources available.	
	Deciding what to conserve comes first. The IUCN Red List uses population size, area, decline rate and probability of extinction to prioritise species, and endemic, keystone, flagship and evolutionarily distinct species receive weight because their loss has wider consequences. This is effective at directing scarce resources, but it favours well-studied vertebrates over invertebrates and plants.	
	Legislation and agreements protect species and habitats. In the UK the Wildlife and Countryside Act, SSSIs, NNRs and Marine Conservation Zones, and in Europe SACs, SPAs and the Habitats Directive, restrict damaging activities; internationally CITES controls trade in threatened species, the IWC regulates whaling, the Ramsar Convention protects wetlands and the Rio Convention on Biological Diversity commits nations to conservation. Their effectiveness depends on enforcement: CITES has reduced trade in ivory and rhino horn but poaching continues where enforcement is weak; the IWC moratorium allowed some whale populations to recover but Japan and Norway continued whaling; designations protect against development but not against pollution or climate change crossing boundaries.	
	In-situ habitat conservation and management are the most fundamental methods because they keep species with the community and abiotic conditions they depend on. Management uses ecological understanding: maintaining plagioclimaxes such as chalk grassland and heathland by grazing or burning, coppicing and pollarding woodland, controlling water levels, light and grazing, designing reserves with the right area, shape, biological corridors and buffer zones, and creating new habitats such as wetlands in worked-out gravel pits. Case studies show it works - coppiced woodland restored dormice and butterflies, wetland creation increased bitterns - but reserves are often small and fragmented and need permanent funding.	
	Ex-situ methods rescue species for which in-situ protection has failed. Captive breeding faces problems of space, diet, breeding triggers, social groups and small gene pools, addressed with studbooks, artificial insemination, embryo transfer and cryopreservation; seed banks such as the Millennium Seed Bank store plant genetic diversity cheaply. The Arabian oryx, California condor and pool frog show captive breeding can bring species back, but only where protected habitat exists for hard or soft release with post-release monitoring, and it is expensive and works for few species.	
	Ecological monitoring - image databases, eDNA, tracking, satellites - measures whether any of this is working and has improved the targeting of effort. Overall, legislation is effective when enforced, habitat management is the most effective and widely applicable method, and captive breeding is a last resort that succeeds only in combination with habitat protection. The Living Planet Index continues to fall, however, which shows that conservation methods are winning individual battles while the drivers of loss - population, consumption, land-use change and climate change - remain unaddressed.	

Do not write outside the box

Question	Marking guidance	Marks
	What makes the difference	
	• The essay is structured by **type of method** and evaluates the effectiveness of each in turn, with named examples for both successes and failures.	
	• It draws on several specification areas (biodiversity, conservation legislation, habitat management, captive breeding, ecological processes, monitoring) - the synoptic breadth Level 5 requires.	
	• The conclusion ranks the methods and then steps back to the bigger picture (Living Planet Index, underlying drivers) - a supported judgement rather than a summary.	
	• A Level 3 essay would describe the same methods accurately but with fewer examples, little evaluation of enforcement or cost, and a conclusion that all methods are important.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Energy flow:** autotrophs and heterotrophs – losses by reflection, transmission, respiration – 100 000 to 1 000 units – few food chains beyond four levels – plant food most productive – grazing animals where crops impossible – pigs on wastes • **Energetics:** intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • **Control of genetics:** asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • **GM:** Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape – advantages – risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity)	25
	• **Sustainability links:** energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	

Do not write outside the box